

Rotating Group Assignments with a Companion Guarantee: Structural Limits, Auditable Screening, and a Hybrid Solver

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Abstract

Repeated group assignment can combine a familiar companion with opportunities to meet other participants. A hard guarantee that each submitter shares a group with at least one named peer, however, can also enable coordinated submissions to force a coalition together. We formalize this directed-list assignment problem with capacities, eligibility states, and two kinds of meeting history. We establish NP-completeness of its general feasibility problem and distinguish structural forcing from global feasibility. An explicit seven-person cycle with a shared outside anchor defeats closure-only screening, including when the closure is large. We introduce a sound, bounded screening procedure that tests whether a candidate core can be separated after relaxing outside obligations; inconclusive solver outcomes remain unresolved. A four-stage heuristic combines construction, repair, simulated annealing, and guarded constraint programming, with independent validation of every retained chart. In a synthetic case study with 257 participants and alternating eligibility states, 10 paired population seeds yield a mean exactly-one-listed-peer rate of 99.64% and a mean increase of +2.46 distinct peers over a capacity-matched random baseline after 16 assignments. These outcomes are conditional on the specified generator and budgets. Published charts also reveal preferences: under a stated public list-size cap, 29.8% of submitted directed entries are logically forced in the reference run. Full traces, an attempt ledger, independent checks, and an interactive playback accompany the study. The procedure does not establish complete resistance to coordinated submissions, feasibility of every admitted input, or psychological and distributive benefits in a real population.

Keywords: group assignment; directed preferences; constraint programming; simulated annealing; strategic submissions; privacy inference

1 Introduction

A rotating lunch assignment has two potentially competing aims: give participants access to a familiar peer and create contact beyond an established circle. We study a concrete operational version of these aims. A participant may submit a list of acceptable companions, and every retained assignment must place that participant with at least one eligible listed peer. Among assignments satisfying this requirement, the procedure discourages additional listed peers and repeated meetings. A declared acceptable companion is an input to an optimization problem; the model does not establish friendship quality, emotional security, or improved belonging.

The guarantee changes the strategic possibilities. A directed cycle of one-name lists forces its members to remain together. Increasing a minimum list length is insufficient when some names become ineligible in a particular rotation, and even two eligible names can force a group through a shared outside anchor. Rejecting every unusual list structure is also unjustified: structural dependence can arise from sincere preferences. An operational system therefore needs explicit proof conditions, an honest account of screening coverage, and a procedure for unresolved submissions.

This paper makes four contributions. First, it specifies the assignment and history objectives precisely and proves a general feasibility hardness result. Second, it develops structural counterexamples and a sound relaxation for detecting forced candidate cores, together with capacity-demand

tests. Third, it documents a working hybrid implementation whose final charts and intermediate stages are saved and independently checkable. Fourth, it evaluates that implementation on a reproducible synthetic study, including coordinated submissions, computational budgets, network structure, and information revealed by released charts. The accompanying website replays actual saved solver stages; it does not claim to optimize a new instance in the browser.

2 Related work

Practical seating optimization has been studied with application-specific objectives and local search [1]. Work on optimal seat arrangement analyzes how preference and seating graph structure affect computational complexity [2]. Our feasible sets differ because a directed acceptable-companion condition is mandatory, groups have exact capacities, eligibility changes across rotations, and past meetings affect the objective. The social golfer problem provides a related repeated-grouping model focused on avoiding repeated pairs [3]. Here repeated listed companionship is sometimes unavoidable: with a fixed list of at most eight names and 16 guaranteed assignments, a submitter must repeat a listed companion.

The feasibility reduction uses the graph-factor framework of Kirkpatrick and Hell [4]. The structural analysis connects to vertex-disjoint directed cycles [5] and partitions of directed graphs under degree conditions [6]. Alon’s directly related *High School Coalitions* note studies the same one-listed-companion condition without the present capacities and changing eligibility states [7]. It already uses the cycle-extension argument and distinguishes the strategic possibilities with two versus three names. We treat these arguments as prior results. Our contribution is their application to state-dependent assignment, an auditable targeted separation test, and the evaluated implementation; we do not claim the general coalition phenomenon as new. The optimization stages use the Metropolis acceptance rule and simulated annealing [8, 9], followed by OR-Tools CP-SAT [10]. Its status distinctions are essential: UNKNOWN supplies neither a feasibility nor an infeasibility certificate [11].

3 Model and objectives

Let V be a set of N participants. Rotation r assigns each participant to exactly one table in $\mathcal{T}_r$, with capacities c_t summing to N . An eligibility state may restrict a table to a grade or other block. Let $L_i \subseteq V \setminus \{i\}$ be participant i ’s submitted list, and let $U = \{i : L_i \neq \emptyset\}$ be the submitters. Filtering the list for rotation r gives $L_i^{(r)}$. If $t_r(i)$ denotes the assigned table, define

$$s_i^{(r)} = \left| L_i^{(r)} \cap \{j : t_r(j) = t_r(i)\} \right|. \quad (1)$$

A valid chart respects eligibility and exact capacities and satisfies

$$s_i^{(r)} \geq 1 \quad (i \in U). \quad (2)$$

A nonempty submitted list that becomes empty after eligibility filtering remains an obligation and makes that state infeasible. A participant who submitted no list has no such obligation. The distinction prevents an implementation from silently converting impossible requests into opt-outs.

Under the defended submission policy, lists are empty or contain at least four distinct names, including two same-grade names. The study generator and attack constructions use at most eight names; the undefended one-name scenario deliberately relaxes the minimum. The generic runner does not enforce a universal eight-name cap on arbitrary custom inputs. The minimum-name rules do not imply global feasibility. List entries are directed: $j \in L_i$ need not imply $i \in L_j$.

For an unordered pair $\{a, b\}$, let $m_{ab}^{(r)}$ count prior incidental meetings, at which neither participant listed the other, and let $\alpha_{ab}^{(r)}$ count prior meetings at which at least one listed the other. Each

meeting contributes to exactly one counter. The annealing objective for rotation r is

$$C_r = \sum_{i \in U} \max(0, s_i^{(r)} - 1) + \lambda \sum_{\{a,b\}: t_r(a)=t_r(b)} \left(5m_{ab}^{(r)} + 2\alpha_{ab}^{(r)}\right), \quad \lambda = 0.1. \quad (3)$$

The first term counts extra listed peers, not merely participants with two or more peers. The second penalizes repeated incidental meetings more strongly than repeated listed companionship. In particular, m is not a count of all meetings with α added again. The CP improvement stage uses a different, explicitly stated surrogate (Section 5.3), followed by an acceptance guard under the original objective.

4 Feasibility and structural limits

4.1 General computational complexity

Theorem 1. *Deciding whether a valid single-rotation chart exists is NP-complete, even with one eligibility block, all capacities equal to three, mutual lists, and every participant submitting a nonempty list.*

Proof. Membership in NP follows by checking a proposed chart. Reduce from P_3 -factor, the NP-complete problem of partitioning the vertices of an undirected graph into three-vertex paths [4]. If the source graph has an isolated vertex or its order is not divisible by three, its answer is NO. Map it to a fixed NO assignment instance: the six vertices of a star $K_{1,5}$, mutual neighbor lists, and two tables of size three. Every leaf must sit with the center, which is impossible.

Otherwise create one participant per source vertex, make every graph edge a mutual list relation, and use tables of size three. Every list is nonempty. A source P_3 -factor supplies a chart in which every participant has a listed tablemate. Conversely, each valid three-person table induces a graph with no isolated vertex. Such a graph is a path or a triangle and contains a spanning P_3 . Selecting one path per table gives the required factor. The construction is polynomial. $\square$

The preprocessing matters because an isolated source vertex would otherwise become a non-submitter with no obligation. The theorem establishes hardness of the general model. It does not establish hardness of every restricted four-name, six-or-seven-seat instance used below.

4.2 Closed sets and forcing

Definition 2. A set $S \subseteq V$ is *guarantee-closed* in a fixed state if every $i \in S \cap U$ has at least one eligible listed peer in S . It is *self-contained* if every eligible name of each such i lies in S . A self-contained set is *structurally coercive* if it admits no partition into two guarantee-closed parts each containing a submitter.

Guarantee-closed means at least one internal acceptable peer, whereas self-contained means all acceptable peers remain inside. The forcing and demand arguments below require the latter condition.

Proposition 3. *All submitters of a self-contained structurally coercive set must share a table in every valid chart.*

Proof. Intersect the set with each occupied table of a valid chart. Self-containment and the guarantee make each nonempty intersection guarantee-closed. If submitters occupy at least two tables, one intersection and the union of the others form a forbidden two-part partition. $\square$

This is a sufficient forcing certificate, not a global feasibility certificate. A noncoercive set may still be impossible to place because of capacities or other participants' obligations. Likewise, the ability to fit a forced group at one table does not certify the remaining population.

Proposition 4. *A self-contained set S with $S \cap U \neq \emptyset$ is not structurally coercive if every vertex of $S \cap U$ has at least three distinct eligible out-neighbors in $S \cap U$.*

Proof. The minimum-outdegree-three theorem gives two vertex-disjoint directed cycles [5]. We use the extension argument described by Alon [7], adapted to the obligated subgraph. Start with the two cycles as separate guarantee-closed sets. Repeatedly attach any remaining obligated vertex with an edge into an existing set. When no attachment is possible, every remaining obligated vertex has all its obligated out-neighbors among the remaining vertices; their union is guarantee-closed and can be merged into the first set. Non-obligated vertices can be assigned freely. Both sets still contain their original cycle. $\square$

Three arbitrary names do not meet this hypothesis if some become ineligible or are outside the obligated subgraph. The implementation applies the shortcut only when all tested vertices have obligations and counts distinct neighbors.

4.3 Coordinated submissions and the shared-anchor counterexample

For odd $g \geq 3$, the directed circulant with edges $i \rightarrow i + 1$ and $i \rightarrow i + 2$ modulo g is structurally coercive. Any directed cycle contains at least $\lceil g/2 \rceil$ vertices: its positive steps sum to a multiple of g and each step is at most two. Two disjoint cycles are therefore impossible. Two nonempty guarantee-closed parts would each contain a directed cycle, giving a contradiction. For even $g \geq 4$, the two parity classes instead provide a split.

Eligibility permits this construction despite a four-name minimum. Seven same-grade participants can list their next two coalition members and two participants from the other grade. In a same-grade rotation the last two names disappear, forcing the seven together. A related six-person attack uses an odd five-person core and a sixth participant dependent on it.

Proposition 5. *Let a coalition $C = \{v_0, \dots, v_{g-1}\}$ have effective lists $L_{v_i} = \{v_{i+1 \bmod g}, h\}$ for a common outside anchor $h \notin C$. Every valid chart seats all members of C together, regardless of h 's list.*

Proof. A table containing a coalition member but not h must contain that member's successor and, inductively, the entire cycle. If the coalition occupied two tables, at least one would exclude h , so that table would already contain the whole coalition, a contradiction. $\square$

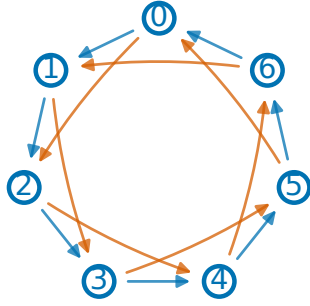
For $g = 7$ and maximum capacity seven, the coalition captures a full table and h sits elsewhere. Adding two opposite-grade names to each list satisfies the four-name rule while preserving the same-grade argument. A successor closure that follows h 's outgoing names can become very large. Even an unbounded test of that larger closure can find a split separating the intact coalition from the background without separating any coalition members. Closure size alone is therefore not the complete problem.

4.4 A sound targeted-core relaxation

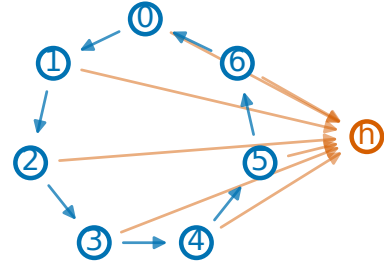
Choose a candidate $C \subseteq U$ and let $B = (\bigcup_{i \in C} L_i) \setminus C$. On $C \cup B$, retain every original outgoing name of each member of C , while removing all obligations of B . Test whether this relaxed system has a guarantee-closed bipartition with a member of C on each side.

Proposition 6. *If no such relaxed bipartition exists, every valid original chart places all of C together.*

Proof. Any original chart separating two core members induces a partition of $C \cup B$ by tables. Every core member retains its actual supporting edge in this restriction, since all its outgoing names were included. Outside vertices have no obligations in the relaxation. Merge the table parts into two groups, retaining a core member in each. The result is a relaxed guarantee-closed bipartition, contradicting the test outcome. $\square$

a Odd seven-member graph

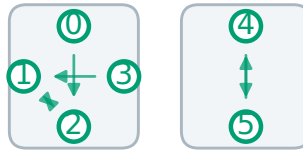
Edges +1 (blue), +2 (orange) mod 7
Directed cycles need at least 4 members;
two disjoint cycles cannot fit.

b Shared outside anchor

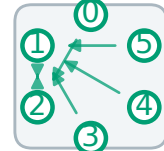
Same-grade names: successor and h
The cycle plus h needs 8 seats;
maximum table size is 7.

c 3+3 partition

Mean cluster = 3

d 4+2 partition

Mean cluster = 10/3

e 6 partition

Mean cluster = 6

Omission star: 0 omits 1; all others omit 0. Each lists the other four.
Bottom arrows show one listed peer per member. All three partitions are feasible.
Mean cluster is weighted by members: the sum of squared part sizes, divided by 6.

Figure 1: Structural examples. The odd circulant and shared-anchor cycle force their cores together. In the six-person omission-star wiring, a feasible partition can be 3 + 3, 4 + 2, or 6. A lower bound of three on member-weighted cluster size does not force exactly two triples or prevent an intact assignment.

A found split clears this necessary test only; it need not extend to a valid chart with the original outside obligations and physical capacities. Outside anchors are not themselves returned merely because they occur in a forcing certificate for the core.

The implementation enumerates successor closures of at most 64 vertices and strongly connected cores of sizes 2–12 in the original directed graph and in graphs formed by deleting one vertex. Candidates are deduplicated, sorted by size and identifiers, and limited to 256 targeted cores. Deleted-anchor edges are restored before testing. Relaxations with at most 20 vertices are exhaustively checked; larger ones use CP-SAT with a deterministic-time budget of 10, one worker, and seed zero. Each candidate receives one of three outcomes: proved separable under the test, proved coercive, or unresolved. UNKNOWN is unresolved and never triggers an automatic coercion return. Coverage limits and unexamined candidate counts are recorded. This candidate family is intentionally bounded and is not a complete anti-capture algorithm.

4.5 Demand and eligibility-specific capacities

Proposition 7. *Consider a self-contained core with n_1 submitters and n_0 nonsubmitters, together with submitting followers whose nonempty eligible lists lie entirely in the core. At most $n_0 + \lfloor n_1/2 \rfloor$ tables can contain any of these participants in a valid chart. Their total number must not exceed the sum of that many largest capacities in their admissible eligibility pool.*

Proof. Every occupied follower table contains a core member. A table with no nonsubmitting core member requires at least two submitting core members, because a submitting core member must find a core peer. Tables with a nonsubmitting core member account for at most n_0 tables; all remaining tables account for at most $\lfloor n_1/2 \rfloor$. The capacity bound follows. $\square$

For example, five mutually linked core submitters and ten followers require 15 seats but can use at most two seven-seat tables. A grade-specific example has 13 senior core members and 29 followers, demanding 42 seats. Their six largest senior tables supply only 41 seats; using the combined grade pools would incorrectly supply 42. If the appropriate pool is unavailable, the implementation reports an unresolved check rather than borrowing another grade’s capacities.

Before generating a year, the implementation also attempts a hard-constraint feasibility solve for each distinct eligibility state. FEASIBLE or OPTIMAL supplies a witness; INFEASIBLE proves impossibility; UNKNOWN leaves the question open. A subsequently constructed and independently validated chart establishes feasibility for that state even if the preliminary solve was inconclusive.

5 Hybrid solution method

Screen-enabled inputs with certified returns, unresolved screening checks, or invalid lists require review before scheduling. The procedure does not invent substitute preferences. Attack scenarios with screening disabled are experiments on the guarantee’s strategic consequences. The screened reference scenario explicitly replaces a returned six-person attack by a specified six-person star; it is one modeled resubmission, not an optimal attacker response.

5.1 Construction and repair

Construction prioritizes participants with few supporting options, forms mutually compatible pods when possible, and merges pods using compatibility and history scores without exceeding the minimum capacity. Packing failures can require singleton placement. The constructed chart has correct capacities but need not satisfy every companion obligation.

Repair uses directed exchanges and compound moves that close supporting dependencies. It attempts to reduce the unsatisfied set without orphaning a previously supported participant; a compound repair is kept only when the prescribed violation and cost checks succeed. A failed local repair does not imply infeasibility.

5.2 Closure-preserving simulated annealing

Let ν count violated companion obligations. The search uses the integer energy

$$E = 10C_r + 10000\nu. \tag{4}$$

The finite violation penalty is not itself a proof that a returned solution satisfies the guarantee. The acceptance filter separately requires $\Delta\nu \leq 0$, and the best assignment is retained lexicographically by violations and cost. Once $\nu = 0$, accepted moves cannot break the guarantee. Before feasibility is reached, the total violation count can remain unchanged even if the identities of unsatisfied participants change.

The proposal mix uses uniform swaps, directed moves, and two-table group moves with probabilities $1/4$, $1/4$, and $1/2$. Group moves close supporting dependencies, balance table sizes, and retry

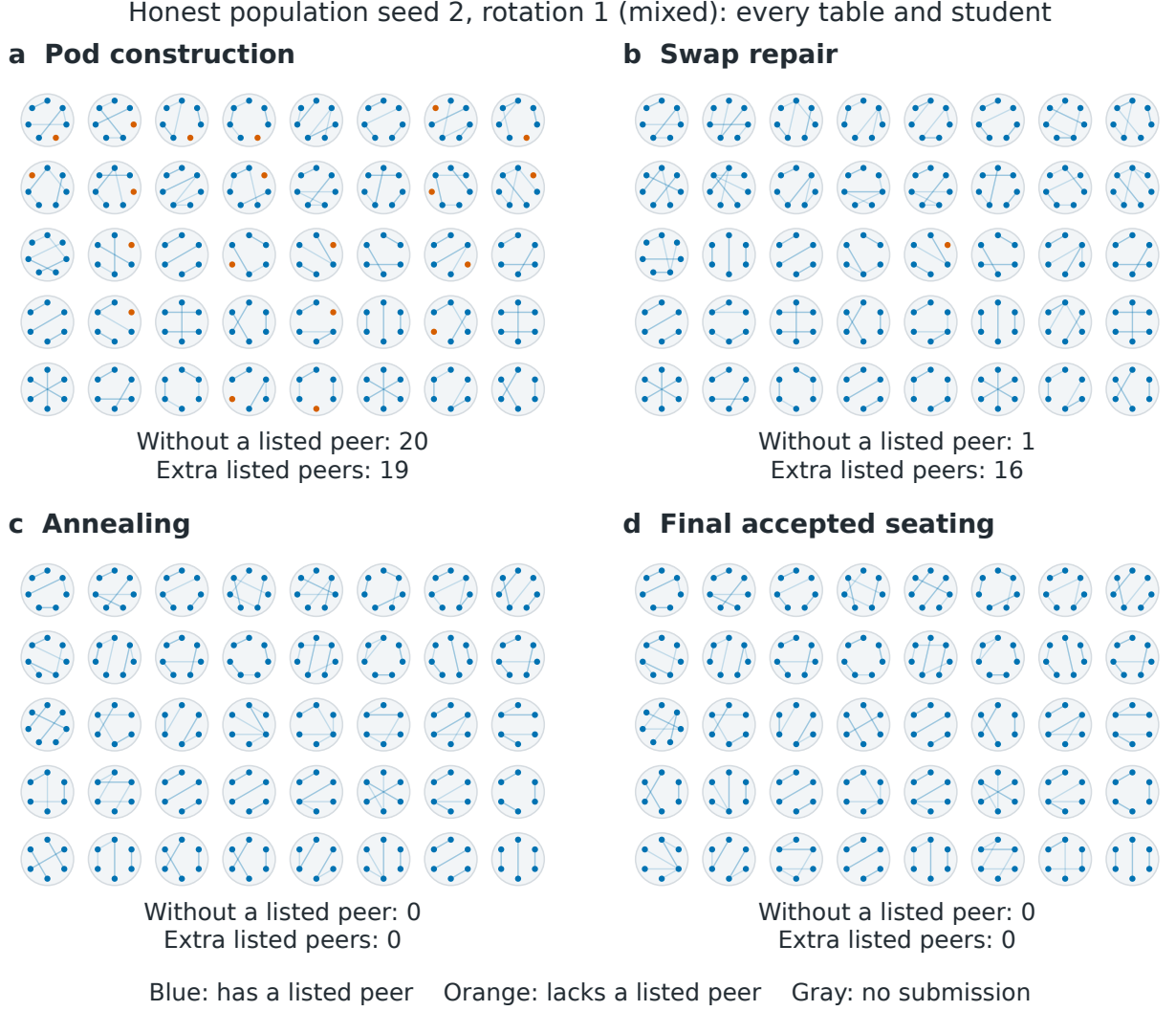


Figure 2: Actual optimization stages for honest population seed 2, rotation 1 (mixed). The four saved stage endpoints are actual assignments. Construction and repair can remain invalid; only a final chart that passes all hard-constraint checks is released.

bounded padding choices. During infeasible search, targeted proposals occur every eight iterations and compound repair attempts every 64. Costs are recomputed on affected tables, including all historical pair weights. A proposal passing the hard acceptance filter is accepted with probability $\min(1, \exp(-\Delta E/T))$. Temperature decreases from 25 to 0.2 over the specified budget, normally 300,000 iterations. This is a finite heuristic; no asymptotic convergence claim is used.

5.3 Constraint-programming improvement and guard

For participant i and eligible table t , binary x_{it} denotes assignment. Exact-one constraints and $\sum_i x_{it} = c_t$ enforce placement and capacities. Supporting indicators satisfy

$$y_{it} \leq x_{it}, \quad y_{it} \leq \sum_{j \in L_i^{(r)}} x_{jt}, \quad \sum_t y_{it} \geq 1 \quad (i \in U). \quad (5)$$

For each submitter, binary z_i marks two or more listed peers at the assigned table. With $\ell_i = |L_i^{(r)}|$, the formulation includes

$$\sum_{j \in L_i^{(r)}} x_{jt} \leq 1 + (\ell_i - 1)z_i + \ell_i(1 - x_{it}). \quad (6)$$

Let P_{r-1} contain the unordered pairs sharing a table in the immediately preceding rotation. For these pairs only, $q_{ab} \geq x_{at} + x_{bt} - 1$ over tables marks a repeated meeting. The coefficient uses each pair’s full history, but pairs absent from the preceding chart are omitted from this surrogate. CP-SAT minimizes

$$300 \sum_{i \in U} z_i + 100 \sum_{\{a,b\} \in P_{r-1}} (5m_{ab} + 2\alpha_{ab})q_{ab}. \quad (7)$$

This objective counts surplus-affected participants rather than all extra peers. Assignment and auxiliary hints are derived consistently from the incumbent, but an invalid incumbent does not become a valid hint merely by being supplied to the solver. The default improvement budget is 3.5 deterministic-time units with eight workers.

A CP candidate replaces a feasible annealing incumbent only if it remains feasible and does not worsen the original cost. If the incumbent is invalid, feasibility takes priority. If annealing remains infeasible, it permits up to six repair-and-anneal retries, each with $\max(I/2, 20000)$ iterations for initial budget I . If the guarded result remains infeasible, a final CP attempt uses $6 \max(b, 1)$ deterministic-time units for improvement budget b ; if it cannot obtain a valid chart, it records an unsuccessful attempt instead of releasing the chart. Every accepted final chart is checked for unique placement, exact capacities, eligibility, and all submitted obligations. Construction, repair, annealing, and final assignments, their costs, and their elapsed times are retained for inspection.

6 Information revealed by published charts

A public chart can reveal private list entries even when the lists themselves are hidden. We analyze an observer who knows that participant i submitted a fixed list, knows a public upper bound K , knows the guarantee was honored, and observes all published charts. Let $T_i^{(r)}$ be the set of i ’s tablemates in rotation r . Every candidate list A must satisfy

$$|A| \leq K, \quad A \cap T_i^{(r)} \neq \emptyset \quad \text{for all observed } r. \quad (8)$$

A name v is forced if excluding v makes these hitting-set constraints impossible under the cap. An exact dynamic program over covered-rotation bitmasks computes minimum hitting-set sizes. An independent CP-SAT set-cover formulation checks the resulting exclusion certificates.

The production inference uses the public cap, not the private true list length. The assumed-public submitter identities are obtained from the input. Actual positive list entries are consulted only afterward to score correctness and the fraction of directed entries recovered. Recovering every true positive entry does not generally determine the entire list: additional names may still fit below the public cap. A list is uniquely determined by forced positives only when those positives fill the public cap, or fill an independently known exact list length. Exhausting the possible peer universe is another sufficient uniqueness condition; this degenerate case does not arise in the study. The output distinguishes complete true-positive recovery from uniquely determined lists.

Under this observer model, no individual name is forced in the first K rotations when every observed table has at least three members. After excluding any one name, each tablemate set remains nonempty; choosing one remaining peer per rotation gives a hitting set of size at most $R \leq K$. This explains the initial flat portion of the leakage curve. It does not mean that charts reveal no information: they already restrict the set of possible lists even without forcing an individual entry.

This observer model omits some possible side information. Additional eligibility or input-rule knowledge can strengthen deductions; uncertainty about submission or compliance can weaken them. These calculations are therefore conditional logical implications under stated knowledge, not a universal privacy-loss metric. The system provides no formal privacy guarantee, and the study does not evaluate a privacy-preserving replacement.

Table 1: Reference configuration. Complete generator, submission, and solver configurations are retained in every trace.

Parameter	Value
Population	257 (132 grade 11; 125 grade 12)
Tables	17×7 and 23×6
Rotations	16, alternating mixed and same grade
Defended study lists	None or 4–8 names; at least 2 same grade
Reference nonsubmitters	0
Popularity / within-grade weight	$\sigma = 0.6 / 9$
Reciprocity target / realized	0.5 / 0.490
Realized within-grade fraction	0.904
Reference communities	$\mu = 0, \omega = 0$
Annealing budget / temperature	300,000 iterations / $25 \rightarrow 0.2$ (integer energy)
CP budget / workers	3.5 deterministic-time units / 8
Pre-feasibility budget	20.0 deterministic-time units per state
Runtime versions	Python 3.14.7, NumPy 2.5.2, OR-Tools 9.15.6755

7 Experimental design

7.1 Synthetic population and admission policy

The reference population has 257 participants: 132 in grade 11 and 125 in grade 12. Forty tables contain 17 groups of seven and 23 groups of six. Same-grade rotations allocate twelve seven-seat and eight six-seat tables to grade 11, and five seven-seat and fifteen six-seat tables to grade 12. Sixteen rotations alternate mixed-grade and same-grade states, starting mixed.

A directed synthetic network targets up to eight true nominations per participant, using lognormal popularity, a within-grade weight, and reciprocal-edge generation. The reference regime reaches eight names; small latent groups at $\mu = 1$ can yield shorter lists. Optional latent groups alter within-group preference and overlap. These parameters are scenario assumptions, not estimates from a school network. A six-person mutual clique is planted for attack comparisons, within one junior latent group when groups are enabled. The construction preserves within-group isolation at $\mu = 1, \omega = 0$. The shared-anchor experiment adds a deterministic seventh coalition member and changes submitted lists while retaining the underlying population. Honest participants whose generated lists fail the admission rule become nonsubmitters; names are not padded in this study. Thus the guarantee’s denominator is the admitted submitter set, and admission exclusions must be reported separately from seating violations. In the retained study every participant submitted an accepted list and there were no rule-based exclusions; this study therefore supplies no outcomes for actually excluded participants.

7.2 Scenarios and structural attack score

The six website reference scenarios share population seed seven: honest submission; an undefended one-name coalition cycle; a six-person four-of-five omission star; a six-person same-grade attack; the seven-person shared-anchor attack; and the six-person attack followed by the specified screened resubmission. The shared-anchor charts are deliberately generated with screening disabled, while a diagnostic records what the revised screen would return.

For a coalition of size q , the member-weighted cluster size is

$$H = \frac{1}{q} \sum_G |G \cap C|^2. \quad (9)$$

An intact count records rotations with all coalition members together. For $q = 6$, all six together may share a seven-seat table with an outsider; this is distinct from capturing a full table. The seven-member coalition fills a maximum-size table when intact.

In the omission-star wiring, every member lists four of the other five, and no one lists the central omitted member. Its feasible internal partition patterns are $3 + 3$, $4 + 2$, and 6 , with $H = 3$, $10/3$, and 6 . The minimum $H = 3$ is a structural lower bound on any valid outcome, not a prediction that every outcome has two triples. Exhaustive enumeration of 5^5 wirings after fixing one omitted name gives the best lower bound 3 among that restricted internal six-person family. It does not optimize larger, externally anchored, or adaptive coalitions.

7.3 Paired random baseline and metrics

Each proposed schedule is compared with independently randomized seat permutations using the same capacities, grade restrictions, and population. For a uniform random rotation in a block of size n , a particular pair meets with probability

$$q_r = \frac{\sum_t c_t(c_t - 1)}{n(n - 1)}. \quad (10)$$

Pairs in different blocks have probability zero. Independence across randomized rotations gives expected distinct peers by summing $1 - \prod_r (1 - q_{ab,r})$ over peers. It does not require independence between different pairs within one rotation. For a participant with k_i eligible names at a table of size c , random companion coverage is

$$1 - \frac{\binom{n-1-k_i}{c-1}}{\binom{n-1}{c-1}}, \quad (11)$$

averaged over table sizes with seat-share weights and over submitters. In the reference population, expected random coverage is 16.05% in mixed rotations and 27.65% in same-grade rotations. Expected final distinct peers are 72.29 for the alternating schedule; the all-mixed value 74.72 is a different design and is not the applicable comparator.

Coverage and exactly-one percentages use submitters as the denominator. Distinct-peer means use all 257 participants. Repeat-pair counts count previously encountered unordered pairs in the current chart; they are not the weighted history cost in Equation (3). Coalition metrics use the actual scenario size. Distributional checks report individual distinct-peer ranges and nomination-degree quartiles, without interpreting those associations as causal effects.

7.4 Study plan, evidence, and uncertainty

The primary comparison uses population seeds 1–10, each with honest and omission-star scenarios at the default budget and 16 rotations. Four additional seed-seven scenarios supply the remaining website references. Sensitivity runs vary annealing budgets to 100,000 and 900,000 iterations, CP budgets to 0.1 and 1.0 deterministic-time units, horizons to 5 and 32 rotations, and latent-group settings to $(\mu, \omega) = (0.6, 0.3)$ and $(1, 0)$. Each sensitivity changes its named setting relative to the seed-seven honest reference; it is not a factorial exploration. Population seeds and solver seeds are separate recorded fields.

Every attempted run is written to an atomic ledger before execution. Successful rows retain complete chart traces, effective inputs, configuration, source-file hashes, dependency versions, the full Git commit, and dirty-tree status. Interrupted, invalid, failed, and unresolved attempts remain visible. Resumption requires matching effective source and configuration identity and verified trace hashes. Table 2 reports actual attempts, including interruptions. Historical summary-only results are retained separately and are excluded from the new numerical tables.

Population seeds, rather than students or rotations, are the independent units for paired comparisons. Reported ranges and a 95% paired-population bootstrap percentile interval (20,000 resamples, PCG64 seed 20260908, linear sample quantiles) describe variation under this generator, conditional on the budgets and fixed study design; they are not uncertainty intervals for a real school population. Runtime measurements were collected with concurrent experiments on a shared machine and are not isolated hardware benchmarks.

Table 2: Attempt-level evidence inventory. Interrupted processes remain in the denominator after resumption; they do not establish either model infeasibility or solver failure. Other terminal outcomes, if any, are detailed in the machine-readable ledger.

Experiment family	Attempts	Success	Interrupted	Other
Primary population pairs	21	20	1	0
Additional reference attacks	5	4	1	0
Annealing sensitivity	2	2	0	0
CP sensitivity	2	2	0	0
Horizon sensitivity	2	2	0	0
Community sensitivity	2	2	0	0

8 Results

8.1 Guarantee satisfaction and contact diversity

All successful charts included in the results pass independent checks of placement, capacities, eligibility, and the companion guarantee. This is a statement about retained, validated outputs; it is not evidence that every rule-compliant input is feasible or that every attempt succeeds. Across 10 honest population seeds, mean exactly-one coverage is 99.64%, with population means ranging from 99.44% to 99.73%. The paired change in final distinct peers averages +2.46, ranging from +2.13 to +3.19; the conditional 95% bootstrap interval for the mean is [2.26, 2.70].

For the seed-seven honest reference, mean exactly-one coverage is 99.68%, and its minimum rotation value is 98.44%. Mean final distinct peers are 74.44 under the proposed schedule and 72.31 under the paired random schedule. Figure 3 shows the full trajectory, while Figure 4 shows population-level variation. The proposed schedule has more repeated pairs than its random comparator in 3 reference rotations (3, 5, 7). An aggregate diversity improvement therefore does not imply a per-rotation dominance result, and exactly-one coverage is an optimization statistic rather than a measure of social benefit.

8.2 Coordinated submissions and screening

The undefended cycle remains intact in 16 reference rotations. The omission star is intact in 0 rotations and has mean cluster size 3.02; observed patterns are 3+3 in 15 rotations, 4+2 in 1 rotation. These observations are compatible with all three feasible structural patterns in Figure 1.

The six-person stratified attack remains intact in 8 same-grade and 0 mixed-grade rotations. The seven-person shared-anchor attack remains intact in 8 same-grade rotations, consistent with Proposition 5. The revised diagnostic returns 7 coalition members; it does not return the common anchor solely because it supports the proof. The screened six-person resubmission has mean cluster size 3.02 and is intact in 0 rotations. This one response illustrates changed behavior after a certified return, not complete strategic resistance. The shared-anchor run also has fewer room-wide distinct

Table 3: Six verified seed-seven reference scenarios. Minimum guarantee and mean exactly-one rates are percentages of submitters; distinct contacts average over all participants. Intact and cluster values use the actual coalition size q .

Scenario	q	≥ 1	Exactly 1	Contacts	Intact	H
Honest	—	100	99.68	74.44	—	—
One-name cycle	6	100	99.59	72.78	16/16	6.00
Omission star	6	100	98.30	74.00	0/16	3.02
Stratified attack	6	100	98.69	73.33	8/16	3.79
Shared-anchor attack	7	100	99.25	69.54	8/16	4.71
Screened resubmission	6	100	98.30	74.00	0/16	3.02

Table 4: Primary population comparisons at the default budget. Exactly-one percentages, distinct contacts (proposed and random), their paired difference, forced directed list entries, and the omission-star mean cluster size are computed from full traces.

Seed	Exactly 1	Contacts	Random	Δ	Forced	Star H
1	99.61	74.74	71.55	+3.19	586	3.00
2	99.59	74.70	72.25	+2.45	622	3.02
3	99.61	75.04	71.95	+3.09	611	3.00
4	99.44	74.70	72.33	+2.37	625	3.02
5	99.68	74.75	72.58	+2.17	637	3.00
6	99.73	74.63	72.39	+2.24	625	3.00
7	99.68	74.44	72.31	+2.13	613	3.02
8	99.71	74.23	72.04	+2.19	629	3.04
9	99.68	74.39	72.08	+2.31	580	3.00
10	99.68	74.58	72.09	+2.49	598	3.00

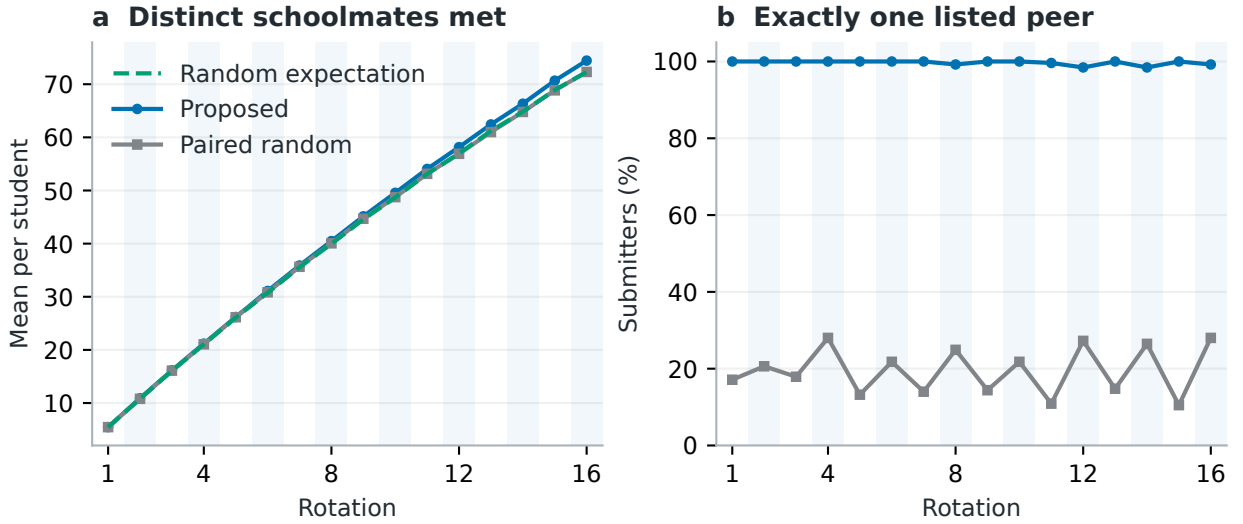


Figure 3: Seed-seven honest reference, using actual saved charts. Contact diversity is compared with the capacity- and eligibility-matched random schedule and its analytic expectation. Pale bands denote same-grade rotations. Companion statistics use submitters; contact diversity uses the whole population.

contacts than the common random baseline, despite satisfying every hard companion constraint. The honest-regime diversity gain therefore does not extend to every tested submission regime.

8.3 Budget and network sensitivity

Table 6 and Figure 5 vary computational budgets, horizon, and latent network structure. A larger budget changes a finite search opportunity; it need not monotonically improve every downstream-year metric. The $\mu = 1, \omega = 0$ latent-group setting has exactly-one coverage 89.88%, compared with 99.68% in the reference. Network concentration can therefore substantially change the apparent ability to minimize extras. At $\mu = 1$, smaller groups also shorten realized lists (mean 7.07 names rather than eight), so this comparison changes both network concentration and the list-length distribution. It is not an isolated causal estimate of community structure. One-population sensitivity runs identify dependence on assumptions but cannot establish general robustness.

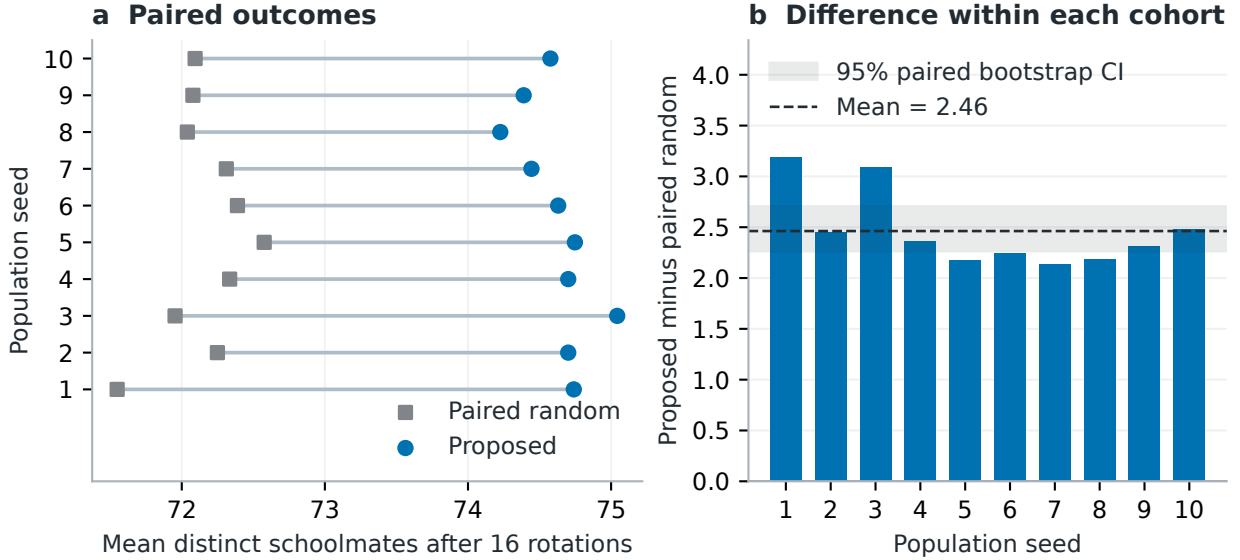


Figure 4: Honest results across independent synthetic population seeds. Pairing preserves the population and eligibility schedule. The line and band in the difference panel show the paired mean and its 95% population-bootstrap interval. They describe this synthetic design, not a sampled school population.

Table 5: Reference diagnostic screening by eligibility state, before any modeled resubmission. Certificates may overlap; they are not counts of distinct students. Target coverage is examined/discovered bounded cores, not recall against all possible attacks. All screens also record closure and capacity checks.

Submission	State	Certificates	Unresolved	Target cores
Honest	mixed	0	0	0/0
Honest	same	0	0	0/0
Shared-anchor attack	mixed	0	0	0/0
Shared-anchor attack	same	1	0	1/1
Initial screened attack	mixed	0	0	0/0
Initial screened attack	same	2	0	0/6

8.4 Privacy implications and distributional observations

In the reference run, the public-cap inference forces 613 of 2056 directed submitted entries (29.8%), affecting 243 participants. Comparison with private synthetic truth confirms 613 forced entries as correct. The inference classifies 0 lists as uniquely determined under the public cap; this count is separate from complete recovery of a shorter true list. Figure 6 displays the accumulation of forced entries and the distribution of distinct contacts.

The 32-rotation sensitivity forces 1691 directed entries (82.2%) and uniquely determines 71 lists under the public cap. The five-rotation probe forces no entries, as predicted by the preceding hitting-set argument. These are one-population horizon comparisons under fixed submissions; they do not imply that every population has the same disclosure rate.

Individual final distinct-peer counts range from 64 to 82. Participants in the lowest and highest incoming-nomination quartiles average 75.78 and 72.48 distinct peers, respectively. These descriptive group means neither demonstrate equal treatment nor establish that nomination degree caused an outcome. They omit other dimensions of fairness, including satisfaction quality and the burden on participants with few acceptable peers.

Table 6: Seed-seven honest sensitivity, with one setting changed at a time. Contact counts and forced entries depend on horizon and should be compared at equal horizons. Nonsubmission includes generated lists rejected by the admission rule; it is distinct from an unmet seating guarantee.

Setting	R	Exactly 1	Contacts	Random	Forced	No list
Reference (16 rotations)	16	99.68	74.44	72.31	613	0
Anneal 100,000	16	99.15	73.82	72.31	554	0
Anneal 900,000	16	99.90	75.13	72.31	688	0
CP budget 0.1	16	99.68	74.54	72.31	602	0
CP budget 1	16	99.66	74.54	72.31	620	0
Horizon 5	5	100.00	26.18	26.15	0	0
Horizon 32	32	99.45	128.26	120.44	1691	0
Groups $\mu = 0.6, \omega = 0.3$	16	99.76	75.03	72.31	544	0
Groups $\mu = 1, \omega = 0$	16	89.88	75.62	72.31	491	0

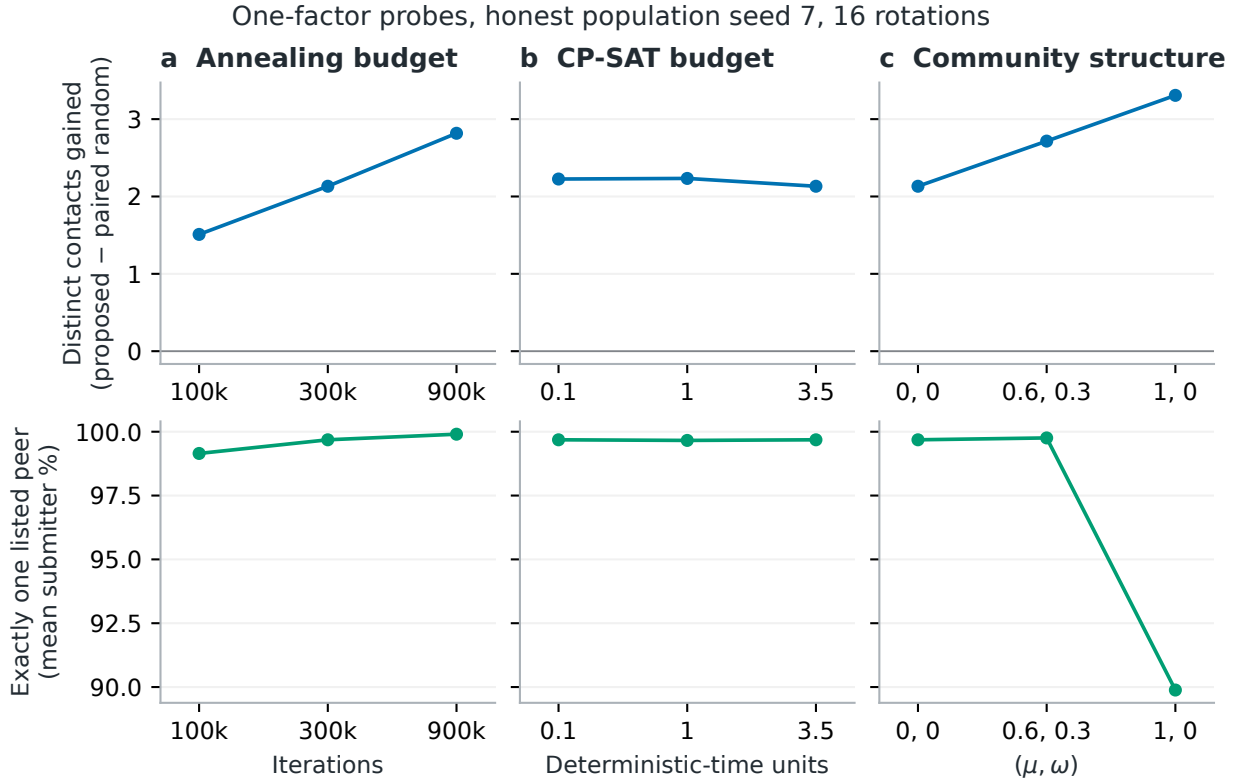


Figure 5: Sensitivity of the seed-seven honest scenario to annealing effort, CP improvement budget, and latent community structure. Other settings use the reference values. Each point represents a single generated population and finite solver run; these are descriptive comparisons.

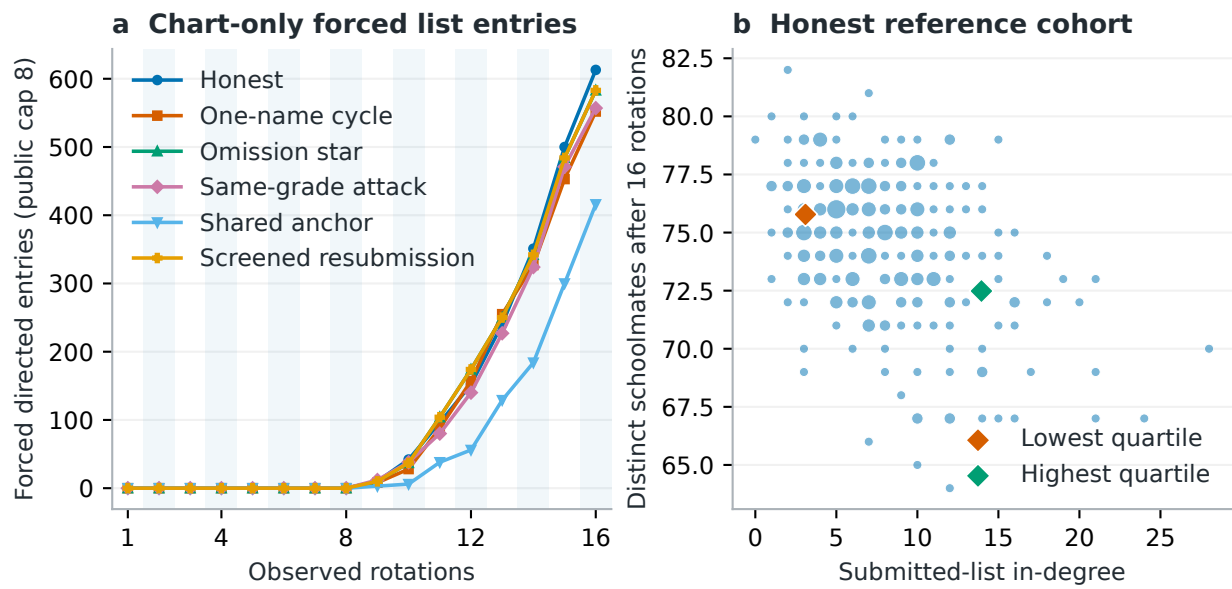


Figure 6: Information implied by successive published charts under the stated public cap across the six reference scenarios (left), and individual contact diversity by incoming nomination degree in the honest reference population (right). Private truth scores inference afterward; it is not supplied to the observer. Pale bands denote same-grade rotations; larger scatter markers combine students at identical coordinates. The degree association is descriptive.

9 Discussion and limitations

The strongest guarantees in this work are local and explicit: every released chart satisfies the stated constraints, and every certified targeted core is forced together under the relaxation’s hypotheses. Neither statement implies that the screen detects every forceable coalition. Single-anchor deletion and a bounded core size target useful counterexamples, but multiple anchors, larger cores, and adaptive submissions remain outside an exhaustive analysis. A screen passage means only that the implemented checks did not certify a problem. A proved-coercive structure does not establish dishonest intent.

Global feasibility remains a distinct question. Minimum list length and two same-grade names do not imply it, and a time-limited unknown solve must not be described as a proof of failure. Where an instance needs review, an operational deployment would need a transparent process involving participants or administrators; this study does not validate such a process. Turning short generated lists into nonsubmission is a documented simulation policy, not evidence that excluding real participants would be acceptable.

The objectives are operational proxies. Fewer extra listed peers and more distinct contacts may be useful design choices, but this study contains no intervention, calibrated friendship network, or measured wellbeing outcome. The synthetic generator, planted coalition, public cap, and fixed budgets restrict external validity. The random baseline measures a particular unconstrained alternative; it does not establish superiority over every feasible heuristic. The study also lacks a full-year optimality certificate and extensive algorithmic ablations.

Publication of charts has a substantive privacy cost in the observer model. Hiding list lengths does not eliminate deductions when a public cap and repeated guarantee are known. The project should not be presented as privacy-preserving. Any future deployment would need a separately specified data policy and a study of mitigations and their effect on the guarantee.

Source and input fingerprints improve auditability, but multiworker optimization and platform differences can still change search trajectories. The attempt ledger and independently validated trace are the evidence for each reported outcome. Historical proposal files describing maximization of two-friend coverage or all-mixed baselines are not the current specification. The original external prototype and collusion-result files described in earlier materials were not available in the repository used for this revision; no claim of reproducing those historical experiments is made.

10 Conclusion

A hard companion guarantee creates both an assignment constraint and a strategic dependency structure. Shared outside anchors show why closure-only reasoning can miss full-table capture. A targeted relaxation yields sound forcing certificates for examined cores, while explicit unresolved states prevent time limits from masquerading as proofs. The implemented hybrid solver produces independently validated synthetic schedules with high exactly-one coverage and modest gains in distinct contacts relative to a matched random baseline. Those gains coexist with detectable preference disclosure and unresolved questions about complete screening, real-world feasibility, and social outcomes.

Declarations

Author and affiliations. Isaac Lin; MIT Lincoln Laboratory; SMU. The institutional labels were supplied by the author. The full institutional name represented by SMU and the final contribution statement must be confirmed before submission.

Funding. Funding information has not yet been confirmed by the author and must be completed before submission.

Competing interests. The author has not yet confirmed a competing-interest declaration; this statement must be completed before submission.

Ethics and participants. The reported experiments use synthetic participants and synthetic preference networks. No human-subject intervention or real student preference data are reported. No ethics approval for a future deployment is asserted.

Data and code availability. Code, traces, verification scripts, and manuscript sources are available at <https://github.com/IsaacLin247/lunch-seating-visual>. The interactive trace playback is at <https://isaacclin247.github.io/lunch-seating-visual/>. The verified attempt manifest identifies the precise effective source and inputs of each experiment.

Use of generative AI. OpenAI Codex assisted with implementation, counterexample checks, manuscript drafting, and figure-generation scripts. Numerical results are calculated from archived program outputs, not generated as prose estimates. The author must review and approve the manuscript, references, and final declarations before journal submission.

Acknowledgments. Independent audit findings motivated the regression suite and the separation of mathematical certificates, heuristic outcomes, and empirical scope.

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A Reproduction and independent verification

The repository pins simulation dependencies in `requirements.txt` and figure dependencies in `requirements-paper.txt`. The exact fresh-study commands and concurrency conditions are documented in `results/STUDY.md`. The saved manifests include population and solver seeds separately, the complete generator and solver configuration, effective input hashes, full Git identity, dirty status, source-file fingerprints, and versions. A source fingerprint covers executable simulation files and simulation requirements, excluding generated figures, manuscript files, and outputs. A dirty checkout is therefore identified explicitly rather than being mislabeled by a short commit alone. The archived study source precedes final hardening of cached-evidence withdrawal, returned-configuration validation, the exact-public-length universe case, and CP feasibility priority over an invalid incumbent. Compatibility checks cover all retained traces. In the seven recorded rotations with invalid annealing endpoints, the first CP call either supplied no candidate or already supplied an accepted lower-cost feasible chart; the priority correction changes no recorded decision.

```
python -m pip install -r requirements-paper.txt
python -m pytest -q
python paper/assemble_evidence.py --retained-source
python paper/make_tables.py
python paper/make_figures.py --retained-source
python results/verification/verify_traces.py \
    results/verified_experiments.json \
    --out results/verification/combined_charts.json
cd article
latexmk -pdf -interaction=nonstopmode article.tex
```

The assembler rejects unfinished inputs for a final release and verifies saved hashes. The trace checker independently recomputes table feasibility, stage costs, rotation statistics, coalition patterns, distributional summaries, and aggregate results. A separate CP formulation checks privacy exclusion claims. The website receives the six completed seed-seven traces, including actual construction, repair, annealing, and final endpoints.

B Regression evidence and audit scope

The regression suite covers the 257-participant shared-anchor witness, restoration of anchor edges during targeted testing, exact senior capacity pools, and unknown solver statuses. Exhaustive seating partitions of 180 small random graphs independently check that every certified core is forced together. These tests validate soundness on the enumerated cases; they do not prove completeness of candidate generation. The star enumeration checks all restricted four-of-five wirings and retains all feasible partition patterns. Privacy tests distinguish a recovered short list from a uniquely determined list under a public cap. Ledger tests preserve interrupted attempts, reject stale or modified traces, and bind resumption to matching source and configuration identity.

The accompanying audit-resolution document maps each follow-up finding to its correction or remaining limitation. Historical source documents and summary-only experiments are preserved as history. Their former objective and numbers are excluded from current claims rather than silently treated as fresh evidence.